

# Complete-knapsack walkthrough: task, trace state, and result

## (a) Frozen task contract

Coins: [1, 2, 5]

Amount: 11

Expected minimum: 3

Strategy: update  $dp[c]$  in  
increasing capacity order

## (b) Problem-specific input and coin affordances

Dynamic Programming

### Coin Change: Minimum Coins

Find the fewest coins needed to make a target amount using given denominations.

Coins: [1, 2, 5]

Amount: 11

Available Coins: 1 - 2 - 5 (no coin active)

## (c) Trace-backed DP state (twelve visible amount states)

DP ARRAY —  $DP[X]$  = MINIMUM COINS FOR AMOUNT X

| x=0 | x=1 | x=2 | x=3 | x=4 | x=5 | x=6 | x=7 | x=8 | x=9 | x=10 | x=11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| 12  | 12  | 12  | 12  | 12  | 12  | 12  | 12  | 12  | 12  | 12   | 12   |

## (d) Navigation and verified terminal result

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Use ← → arrow keys to step through frames

VERIFIED ANSWER VERIFIED

**3**

= 5 + 5 + 1 (3 coins)

## (e) Executable evidence

68 trace frames (0–67)

verified answer = 3

browser audit = 9/9

**ALL NINE CHECKS PASS**